Transport project: Self-assessment checklist

Please use the rubric that you can find on CANVAS to award yourself a mark for the four categories in the assessment rubric you can find on CANVAS. Please provide a justification for the marks you have awarded yourself that explains how your report addresses the assessment criteria in the marking rubric.

PROBLEM: *The student must describe the problem they have worked on and why solving it matters* *9.5* /10

- **I clearly explain the importance of road safety in NI**
- **Give motivation as to why Fermanagh has particular road safety issues**
- **Use statistics, quotations and literature to justify the problems relevance**
- **Linked my work to FCT showing why the problem matters to them**
- **Reference similar approaches made previously, quoting a study done on a similar topic giving a brief summary of their work**

MODELLING: *The student must describe the model that has been used to tackle the problem that they have identified.* *9* /10

- **The model is clearly and correctly described with justification for the choice of Negative binomial distribution due to overdispersion**
- **Discussed assumptions, limitations and extensions**
- **Presented key diagnostics to show the models performance**
- **Clearly stated variables**
- **Included log-likelihood, link function and predictor definitions**

- **Didn't include residual plots or other helpful diagnostics, this was due to lack of space**

INTERPRETATION: *The student must describe the results that they obtained from their modelling study.* *9* /10

- **Explained the reasoning behind why each figure was constructed**
- **Interesting inferences made for all 3 models**
- **Sites and studies referenced to back up my interpretation**
- **Described uncertainty in sparsely populated cells**
- **Connected the results back to FCT**

- **Didn't include confidence intervals or error bars, I am unsure of how I would do this for the model I have made**

PARTNER: *The student describes why their work matters to the community partner*
5 /5

- **My recommendations for policies are specific, actionable and directly tied to the results I found in my model**
- **I clearly explain why the findings matter to FCT**
- **I reference policy documents and FCT's operational context (minibuses and cars used as taxis)**



Queen's University Belfast

School of Mathematics and Physics

SOR3012: Stochastic Processes and Risk

Student Name: Ben Hamilton

Student Number: 40405601

Date: January 3, 2026

Abstract

Road traffic accidents remain a significant concern in Northern Ireland, contributing to fatalities, injuries and economic damage. This study develops a statistical model of road traffic accident patterns and forecasts future collision hotspots using stochastic processes. By analysing historic data, the research identifies important risk factors and high-risk locations. The study's findings provide actionable insight for Fermanagh Community Transport to improve driver safety and minimise financial losses.

1 Introduction

In Northern Ireland, on average one person dies on the roads every week and hundreds of others are injured. Road accidents do not pick sides and have no memory; anyone can become a victim. To quote John O'Dowd, Northern Ireland's Infrastructure Minister: "The devastation each and every road death leaves in its wake knows no boundaries."[1]

1.1 Motivation

Figure 1 shows the deaths per 100,000 people in Northern Ireland by district.[2]

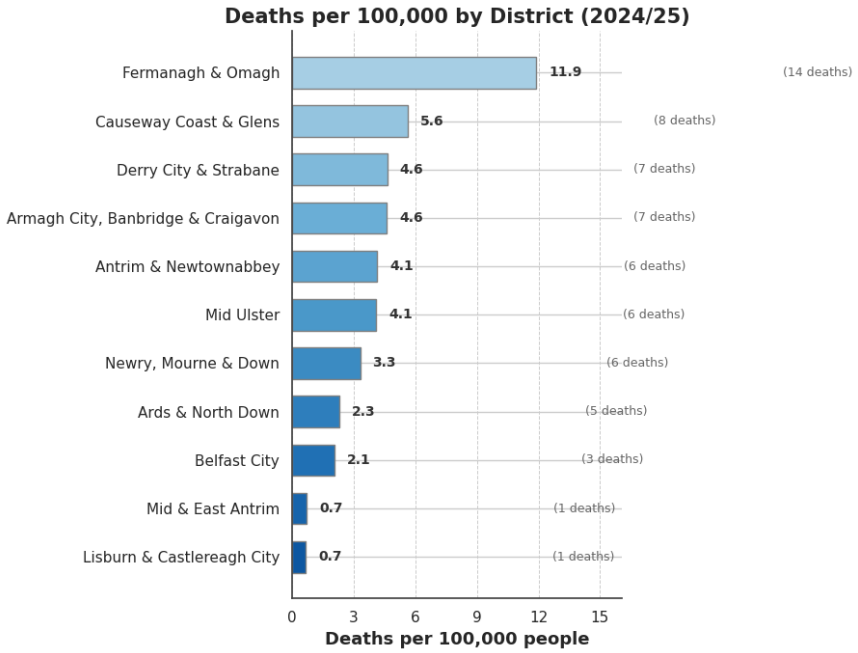


Figure 1: Deaths due to road traffic accidents per 100,000 people in Northern Ireland by district (2024/25).

From the figure we see Fermanagh and Omagh as a clear outlier; its death rate due to traffic accidents is over double the next highest district. Previous studies[3] on collision risk have commonly used Poisson and Negative Binomial models to identify where accidents tend to cluster and how infrastructure and environmental factors contribute to those risks. Our analysis applies to FCT[4] a transport charity working in Fermanagh and Omagh. This aligns with the aims of the NI Road Safety Strategy 2030, which prioritises evidence-based targeting of high-risk locations.[5]

2 Data collection and processing

2.1 Data collection

This study utilises collision data for Fermanagh and Omagh over four years (2021–2024). The data was collected and recorded by the PSNI[6] and is available on Open Data NI,[7] who also provided the surface defects data, for which we used records from 2024.[8] Finally, weather statistics were obtained from WeatherSpark.[9]

[1] For more information. See https://www.irishnews.com/news/northern-ireland/honest-conversations-needed-around-road-safety-infrastructu

[2] Districts are local government administrative areas; Northern Ireland has 11 districts. See https://en.wikipedia.org/wiki/Local_government_in_Northern_Ireland for an overview.

[3] For one influential contribution, see Aguero-Valverde & Jovanis (2006), who showed that fatal and injury crashes form clear spatial clusters strongly influenced by road layout, traffic exposure and local environmental conditions. https://www.researchgate.net/publication/7322037_Spatial_Analysis_of_Fatal_Injury_Crashes_in_Pennsylvania

[4] Fermanagh Community Transport

[5] For more information see https://www.infrastructure-ni.gov.uk/publications/road-safety-strategy-northern-ireland-2030

[6] Police Service of Northern Ireland

[7] https://admin.opendatani.gov.uk/dataset?organization=police-service-of-northern-ireland&tags=injury+collisions

[8] https://admin.opendatani.gov.uk/no/dataset/surface-defects

[9] https://weatherspark.com/

2.2 Data processing

The collision datasets collected from Open Data NI included data corresponding to the policing area, collision severity, and the date and time of the collision. Each dataset contained columns `a_gd1` and `a_gd2`, which corresponded to Easting and Northing coordinates. These were converted to latitude and longitude so the collisions could be plotted as points on a map in Python. This was also applied to the surface defects dataset and was achieved using `pyproj.Transformer.from_crs('EPSG:29903','EPSG:4326', always_xy=True)` to convert Easting/Northing to lon/lat, and `geopandas.GeoDataFrame(..., geometry=gpd.points_from_xy(defects['lon'], defects['lat']), crs='EPSG:4326')` to create the GeoDataFrame. [10]

Several columns in the collision datasets were removed due to being out of scope or severely incomplete (14 out of the 23 columns). For example, the column indicating whether an officer attended the scene was not considered for this study.

2.3 Spatial Framing

To showcase traffic collision hotspots, the study divides Fermanagh and Omagh into 2 km square grid cells and overlays an outline of the study area. The area boundary was obtained as a GeoJSON file from MapIt UK.[11]

To plot the grid lines, I projected the boundary to a suitable projected CRS, generated square grid cells of the chosen size, and retained only cells that intersect the boundary. This was done using `xs = np.arange(minx, maxx + cell_size_m, cell_size_m);` `ys = np.arange(miny, maxy + cell_size_m, cell_size_m)` to generate the grid coordinates, and `gpd.GeoDataFrame({'grid_id': ids, 'geometry': polys}, crs=proj_crs).reset_index(drop=True)` to assemble the grid.

3 Results and discussion

The results are presented in three parts: (i) general observations for Fermanagh and Omagh, (ii) the statistical model for collision hot spots, and (iii) findings interpreted in the context of FCT. Each subsection briefly outlines the data processing and figure construction before presenting the key inferences.

3.1 The general observations of Fermanagh and Omagh

Using the Collision2024 data set we are able to produce a plot that will show at what time of the day each collision occurs. We can find the population of each district using statistics from NISRA[12] We can then use this data to plot the collisions per capita [13] at each hour.

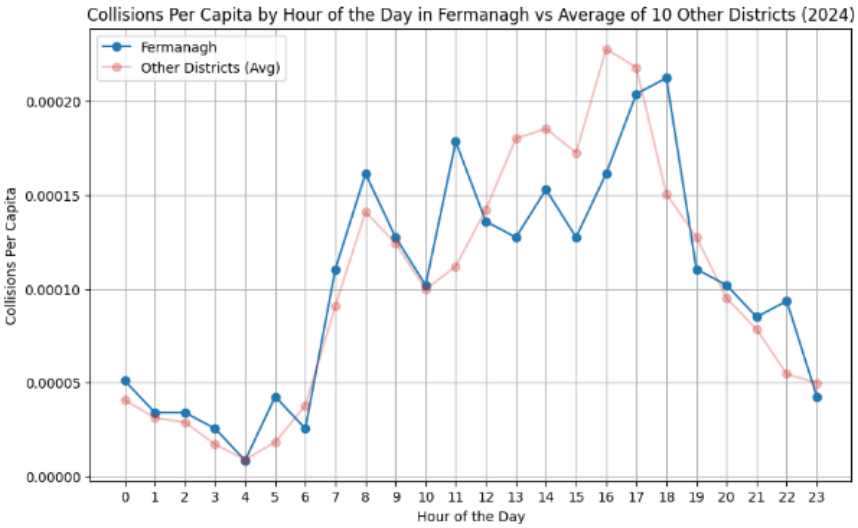


Figure 2: The plot compares hourly collisions per capita in Fermanagh and Omagh (blue) against an average of the other 10 districts (pink) in 2024. Both show clear prominent peaks at 8:00 and 17:00. Fermanagh largely follows the same pattern while differing in magnitude at some hours.

[10] EPSG:29903 is a projected coordinate system called Irish Grid, used for mapping Northern Ireland's onshore areas. EPSG:4326 is the WGS84 geographic coordinate reference system, which uses latitude and longitude.

[11] GeoJSON is an open format for encoding geographic data structures. See https://doc.arcgis.com/en/arcgis-online/reference/geojson.htm for details.

[12] Northern Ireland Statistics and Research Agency. See https://www.nisra.gov.uk/publications/ct0116-usual-resident-population-local-government-district-and-urban-status.

[13] "Collisions per capita" means the number of collisions divided by the population of the area, allowing fair comparison between districts of different sizes.

3.1.1 Inferences

From Figure 2, Fermanagh and Omagh generally follow the same hourly collision pattern as other districts, though two notable differences emerge:

1. **Lower daytime collision rates**: During most working hours, i.e. from 12:00 to 18:00, the collision rate for Fermanagh is lower than that for the rest of the country. This could be due to the fact that the unemployment rate in Fermanagh and Omagh is the highest in the country by a considerable margin[14].

2. **Distinct morning and evening spikes**: There is a spike in Fermanagh and Omagh from 10:00 to 11:00 compared to other districts with another spike from 17:00 to 18:00. As Fermanagh and Omagh is a more rural area this could be due to further travel times to and from work/school. This is supported by Fermanagh and Omagh having the greatest average distance to both dentist and pharmacy[15].

3.2 Statistical model for crash hotspots

Before presenting the model it is important to understand how it was formed and the intentions behind its creation. For modeling accident counts, Poisson regression is a fundamental approach. However, due to overdispersion in the accident data[16], negative binomial regression has been used as a more flexible alternative (Lord & Mannering, 2018)[17].

3.2.1 Model formation

Let Y_i be the total crashes in grid cell i aggregated over 2021–2024 (integer counts).

The collision counts were aggregated to 2 km by 2 km grid cells using the Collision2021–2024 datasets merged with the following predictors:

- **Surface defect count**[18] – capped at the 99th percentile to limit leverage from extreme cells (units: simple count, capped).
- **Nearest town average wind speed** – average wind speed from 2024 for Irvinestown, Omagh, Lisnaskea, Enniskillen and Fintona; each cell inherits the nearest town wind speed (units: kph) and this predictor is standardised to mean zero.
- **Distance to nearest town** – using the 5 towns listed above; transformed as $\log(1 + \text{dist}_i)$ to reduce skew (distance in metres).

These predictors were chosen because they represent key physical and environmental factors known to influence collision likelihood, including road condition, weather exposure, and rural travel distance.

The following Negative Binomial model is used to predict collision counts in each grid cell:

$$Y_i \sim \text{NegBin}(\mu_i, \kappa), \qquad \log(\mu_i) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i}.$$

The predictors are:

- x_{1i}: capped defect count,
- x_{2i}: standardised wind speed,
- x_{3i}: standardised $\log(1 + \text{distance}_i)$.

Under the log link, $\mu_i = \exp(\mathbf{x}_i^\top \beta)$, so coefficients act multiplicatively on the expected crash count.[19]
The Negative Binomial log-likelihood is

$$\ell(\beta, \kappa) = \sum_i \Big[\log \Gamma(Y_i + \kappa) - \log \Gamma(\kappa) - \log(Y_i!) + \kappa \log \kappa - \kappa \log(\kappa + \mu_i) + Y_i \log \mu_i - Y_i \log(\kappa + \mu_i) \Big],$$

with $\mu_i = \exp(\mathbf{x}_i^\top \beta)$. Maximum likelihood estimation provides $\hat{\beta}$, $\hat{\kappa}$, and standard errors.
The model is fitted in Python using `statsmodels`:

```
glm_nb = sm.GLM(y, X_all, family=sm.families.NegativeBinomial()).fit()
print(glm_nb.summary())
```

[14] For further reading see https://share.google/OLeZuWK2omghRL5Xn

[15] For further reading see section 3.0 of the Fermanagh and Omagh area funding report https://share.google/Q9FdEoU1guf7ue8ky

[16] The observed variance in the data is significantly larger than the mean, violating the Poisson assumption that the mean and variance are equal.

[17] See https://www.sciencedirect.com/science/article/pii/S0965856410000376

[18] Surface defects are recorded using a range of defect categories, e.g. pothole, rutted, depressed, cracked, uneven, etc. For more information see https://admin.opendatani.gov.uk/no/dataset/surface-defects

[19] Exponentiating the linear predictor returns the expected number of crashes.

3.2.2 Model Diagnostics and Error Considerations

Diagnostic	Value
Log-likelihood	-1118.9
Deviance	885.84
Pearson χ^2	1.25×10^3
Pseudo R^2 (Cragg–Uhler)	0.354
Observations	856

Table 1: Key diagnostics from the fitted Negative Binomial model.

All predictors are statistically significant at the 5% level, indicating that surface defects, wind speed, and distance to the nearest town each contribute meaningfully to explaining variation in collision counts. The Pearson χ^2 statistic shows residual over-dispersion, and the pseudo-R^2 of 0.354 suggests moderate explanatory power. Predicted values range from 0 to 51 crashes per cell, with greater uncertainty in sparsely populated areas. These results should therefore be interpreted as indicators of relative risk rather than precise forecasts

3.2.3 The Model

The following figure 5 presents a choropleth[20] of expected traffic collisions aggregated to 2 km×2 km grid cells.

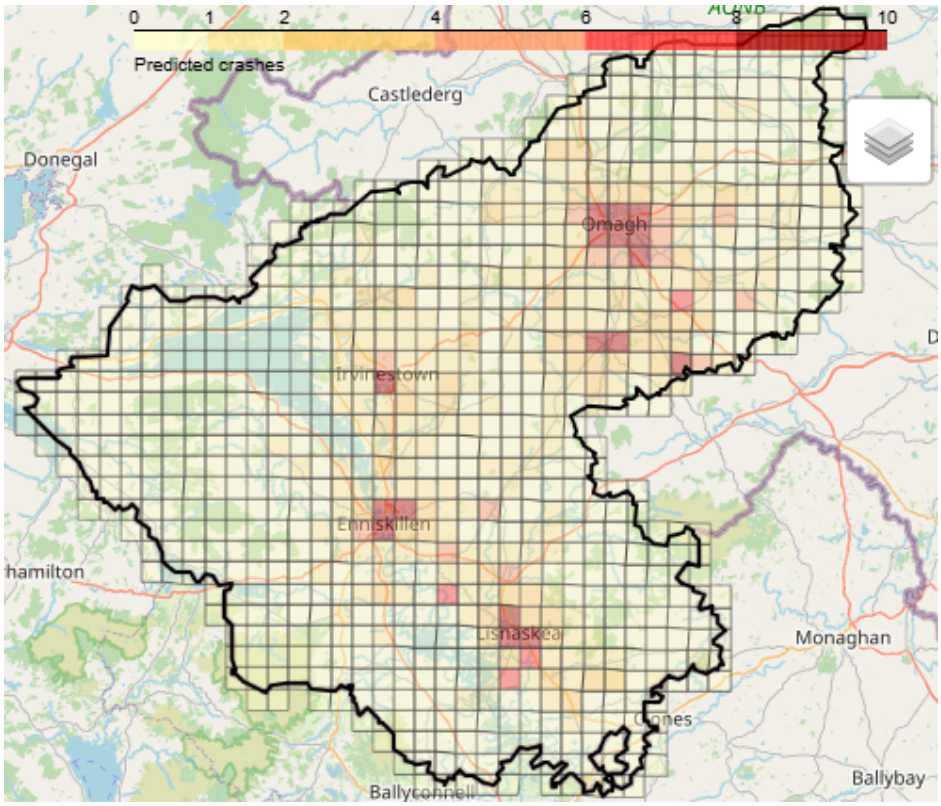


Figure 3: Choropleth of modelled expected crashes per 2 km cell in 2025 from a Negative Binomial GLM using defect counts, nearest-town annual wind speed and distance to nearest town. The darker cells show places with higher expected collision counts

3.2.4 Inferences

From the figure, we observe:

1. **Urban and rural clustering**: The majority of collision hotspots in Fermanagh are in highly populated areas. This is in line with the rest of the UK, where roads in urban areas account for a much higher proportion of road casualties (62%) than their relative traffic level (35%)[21]
2. **Commute-related risks**: Many predicted crashes occur on roads surrounding these populated areas. Since 46% of the district's population live in open countryside[22], a large proportion of residents commute into towns by road, increasing traffic on approach routes and the associated collision risk.

To support the interpretation of these results, the main assumptions of the Negative Binomial model are outlined below.

[20] A map composed of coloured polygons; see `https://plotly.com/python/choropleth-maps/`.
[21] RAC Foundation, Question 8. See `https://www.racfoundation.org/motoring-faqs/safety`.
[22] Fermanagh and Omagh 2030 community plan, Section 2: Fermanagh and Omagh 2030 community plan.

3.2.5 Assumptions

The modelling approach assumes that collision counts within each grid cell are independent, that the relationship between the predictors and expected collisions is adequately described by a log-linear form, and that over-dispersion is captured through the Negative Binomial variance structure. Predictions should be interpreted with uncertainty, particularly in sparsely populated cells where collision counts are low and estimates are less stable.

3.2.6 Limitations

The analysis is limited by the spatial aggregation of collisions, the small sample size, and the absence of additional explanatory factors such as driver behaviour or real-time traffic exposure. These constraints mean that the results should be interpreted as patterns rather than precise forecasts.

3.2.7 Extensions

Future work could build on this model by adding spatial effects and year-to-year trends, incorporating exposure measures such as traffic flow or passenger-kilometres, and tying the predictions more closely to FCT's actual routes to give a clearer picture of operational risk. Similar approaches have been used in transport-safety research, such as the spatio-temporal modelling framework proposed by Li et al. (2019)[23]

3.3 Observations fitted for the context of FCT

Given that FCT is a charity that operates minibuses and cars to provide transport services across Fermanagh, the vehicle CSV file for each year (which corresponds to the collision data) was filtered to two types: `v_type = 18` (car used as a taxi) and `v_type = 19` (minibus). Using these filters, a point-clustering map was created to show where collisions occurred from 2021–2024.

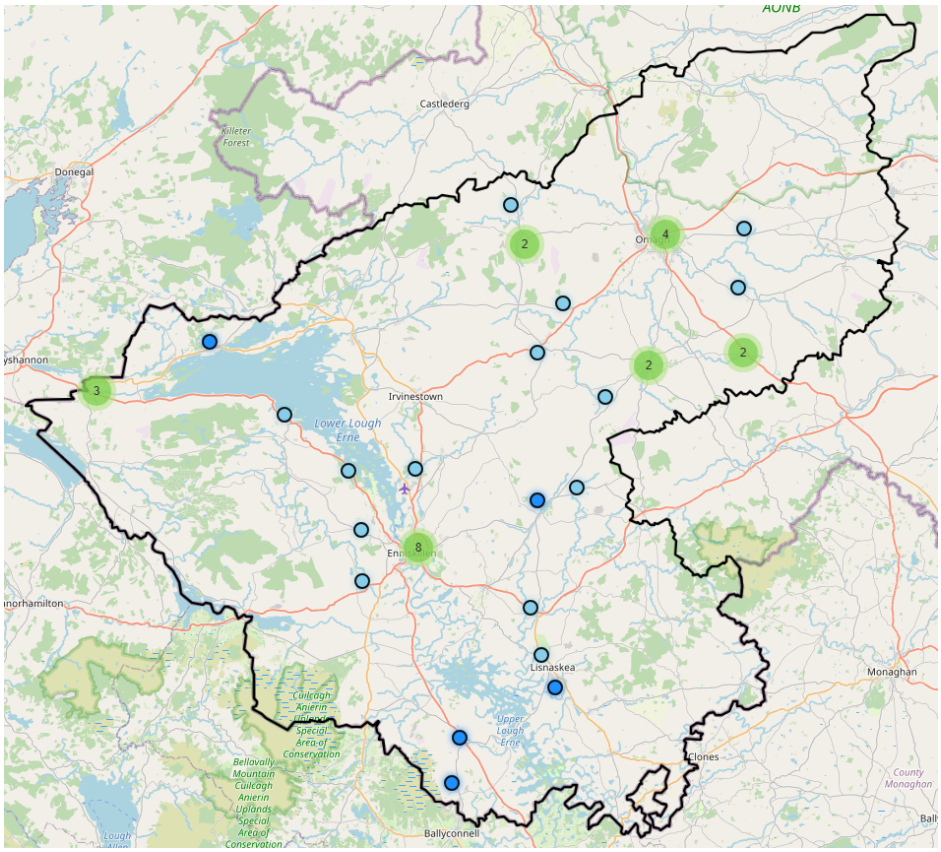


Figure 4: Map of collisions for vehicle types 18 and 19 (2021–2024). Dark blue points represent serious injury collisions; light blue points represent slight injury collisions. Nearby points are clustered for clarity.

Further refining of the dataset can show us where only serious injury collisions take place

[23] See `https://ieeexplore.ieee.org/document/8917408`.

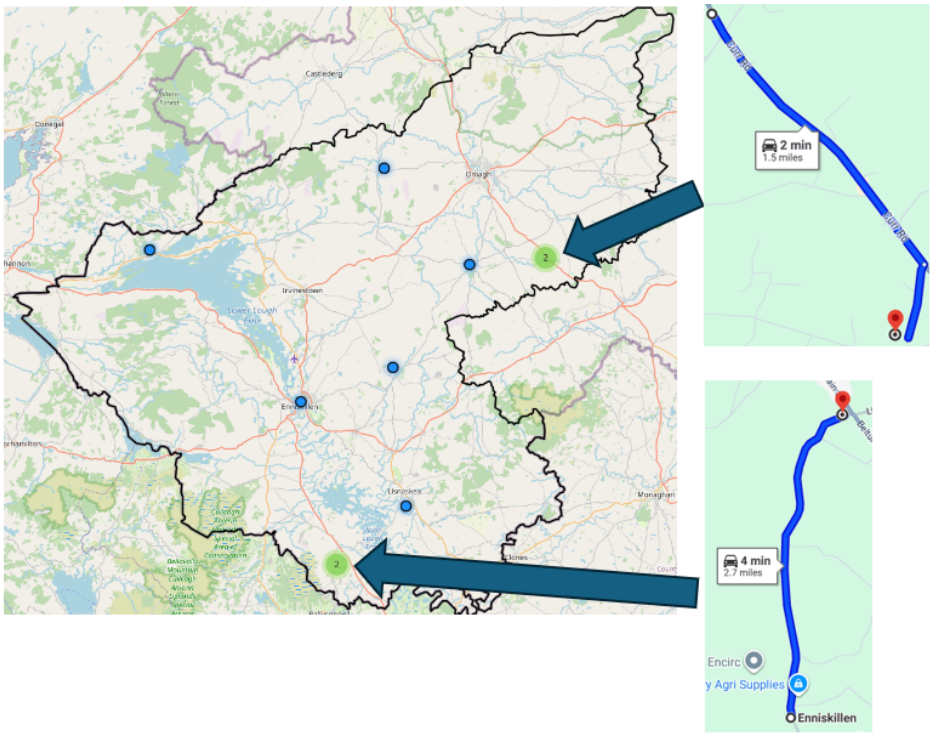


Figure 5: Map of serious injury collisions for vehicle types 18 and 19 (2021–2024). The arrows show the clusters where serious accidents have happened close by

3.3.1 Inferences

From the figures, we observe:

1. **Urban clustering of minor collisions**: For minibuses and cars used as taxis, collisions are still clustered around highly populated places although these collisions are mostly not serious. This will be due to the speed regulations in the centre of populated towns

2. **Rural concentration of serious collisions**: The serious collisions are happening mostly in more rural areas of Fermanagh [24] This will be due to higher speed limits and longer journeys. Further 6 of the 10 collisions occur on the weekend, suggesting higher risk.

4 Conclusion

4.1 Policy recommendations for FCT

The observations presented here can help FCT prioritise interventions, target training, and support evidence-based operational decisions.

1. FCT should place additional emphasis on driver awareness and fatigue management during the morning and evening commuting periods, as longer commute times in Fermanagh extend the effective 'rush-hour' risk window (see Figure 2).

2. Route planning and safety briefings should focus on busy town approaches and main corridors serving population centres, where collision frequency is highest due to high traffic volumes from rural commuting.

3. Serious-injury collisions are most common on rural arterial roads that form key parts of FCT service routes, indicating that these corridors represent the highest-priority environments for risk awareness and safe-driving emphasis.

As FCT is also the delivery partner for MiDAS [25] training in Fermanagh, the findings of this study can provide useful evidence to support and refine that programme. In particular, the assumptions outlined in Section 3.3 highlight the conditions under which collision risk is likely to vary across different road environments. Understanding these patterns can help MiDAS emphasise specific driving contexts where their drivers can be at risk.

[24] The arrows show the Curr road in the North East and the R205 in the South East
[25] For more information. See `https://www.fermanaghcommunitytransport.com/training`

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